Wave Behaviour

Exam code: 0625 & 0972

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Reflection, refraction & diffraction

- All waves, whether transverse or longitudinal, can undergo:
  - **reflection** at a plane surface
  - **refraction** due to a change of speed
  - **diffraction** through a narrow gap
- In optics, a transparent material is called a **medium**
  - When referring to more than one medium these are called **media**
- Angles of light are measured from an imaginary line called the **normal**
  - The normal is always **drawn perpendicular** to the **boundary** between two media

Reflection

- Reflection occurs when:

A wave hits a boundary between two media at a plane surface and does not pass through, but instead stays in the original medium

An example of reflection

An identical image of the tree is seen in the water due to reflection

Help me understand

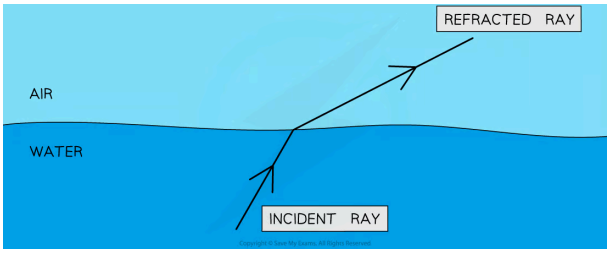
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

change
- This effect is called **refraction** and it occurs when:

A wave passes a boundary between two different transparent media and undergoes a change in speed

- When a wave **refracts**, as well as a change in speed, the wave also undergoes:
  - A change in **wavelength** (but frequency stays the same)
  - A change in **direction**

An example of refraction



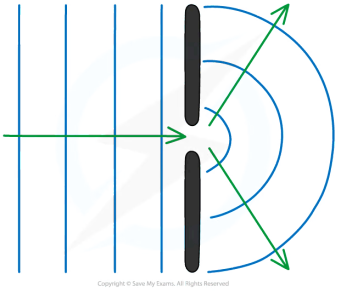
Waves can change direction when moving between materials with different densities

- The direction of the incident and refracted rays are also taken from the **normal** line
- If the waves **slow down**, they will bunch together, causing the wavelength to decrease
  - The waves will also start to turn slightly **towards** the normal
- If the waves **speed up** then they will spread out, causing the wavelength to increase
  - The waves will also turn slightly **away from** the normal

Diffraction

- When waves pass through a narrow gap, **the waves spread out**
- This effect is called **diffraction**

Waves diffracting through a narrow gap



Diffraction: when a wave passes through a narrow gap, it spreads out

Examiner Tips and Tricks

When drawing waves being reflected take care to:

- Make sure that the angle of incidence is equal to the angle of reflection
- Keep the wavelength of the waves the same

Similarly, when waves are diffracted **the wavelength remains constant.**

Refraction is the only wave effect in which the wavelength changes.

Remember:

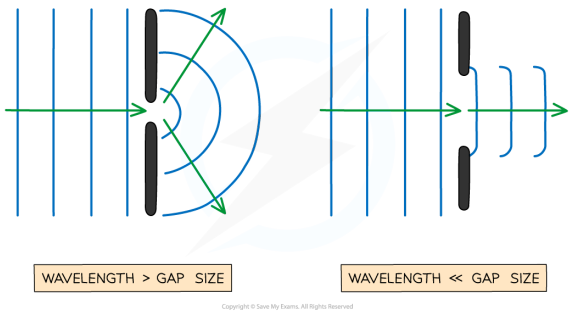
Refraction is the name given to the change in the speed of a wave when it passes from one medium to another. The change in direction is a consequence of this.

Factors affecting diffraction

Extended tier only

- The extent of diffraction depends on the **width** of the gap compared with the **wavelength** of the waves
  - Diffraction is the most prominent when the **width** of the slit is approximately **equal** to the **wavelength**
- As the **gap gets bigger**, the **effect** gradually gets **less pronounced** until, in the case that the gap is very much larger than the wavelength, the waves no longer spread out at all

Effect of gap size on diffraction



The size of the gap (compared to the wavelength) affects how much the waves spread out

- Diffraction can also occur when waves curve around an **edge** or **barrier**
- The waves spread out to fill the gap behind the object
- The extent of this diffraction also depends upon the wavelength of the waves
  - The **greater** the **wavelength** then the greater the **diffraction**

When a wave goes past the edge of a barrier, the waves can curve around it. Shorter wavelengths undergo less diffraction than longer wavelengths

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